

Task 4: Safety, Guardrails & Internal Evaluation



Overview

In this task, your objective is to implement clinical safety guardrails, calibrate empirical confidence thresholds, and perform end-to-end evaluation using the Jupyter notebook:

- `Task4_Safety_Evaluation.ipynb`

While earlier tasks established ingestion, retrieval optimization, and grounded generation, Task 4 focuses on **safety, verification, and defense-in-depth**. The objective is to ensure the clinical assistant reliably refuses out-of-scope questions, detects hallucinated or unsupported claims before they reach clinicians, and produces measurable evaluation metrics to prove system readiness.



Objectives

By completing this task, you will:

1. **Empirically Calibrate Confidence Thresholds:** Analyze retrieval score distributions across answerable vs. unanswerable questions to set data-driven confidence cutoffs.
 2. **Implement an Unsupported-Claim Detector:** Build a secondary safety net to verify that claims in generated recommendations have direct lexical and semantic backing in retrieved evidence.
 3. **Benchmark End-to-End Performance:** Run evaluations on a structured benchmark containing both clinical retrieval queries and deliberate refusal/safety test cases.
 4. **Measure Key Evaluation Metrics:** Compute quantitative evaluation metrics including Retrieval Precision@k and Safety/Refusal Pass Rates.
 5. **Diagnose & Isolate Failure Modes:** Identify whether performance bottlenecks originate from chunking/retrieval or safety threshold calibration.
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Step-by-Step Task Guide

1. Setup & Environment

- Open `Task4_Safety_Evaluation.ipynb`.

- Run the setup cell to rebuild/load the index and prepare evaluation dependencies.

2. Calibrate Retrieval Confidence Threshold

- Run retrieval across known answerable clinical questions and unanswerable/out-of-scope queries.
- Compare the resulting retrieval score ranges to identify the empirical boundary gap.
- Select a data-backed confidence threshold to govern automated refusal behavior.

3. Implement Unsupported-Claim Safety Net

- Implement sentence-level claim extraction and evidence-overlap verification functions.
- Test the detector against supported clinical answers vs. drifted answers containing unsupported specifics (e.g., ungrounded dosages or intervals).
- Verify that drifted claims are caught and flagged by the safety check.

4. Execute Benchmark Evaluation

- Ingest the benchmark evaluation dataset containing retrieval questions and safety/refusal test cases.
- Evaluate each test case for correct retrieval matching (Precision@k) or appropriate refusal triggering.
- Calculate the aggregate Average Precision@k and Safety Correct-Behavior Pass Rate.

5. Diagnose & Document Findings

- Review computed metrics to isolate performance bottlenecks (retrieval vs. threshold calibration).
- Record calibrated thresholds and verified performance metrics for system presentation.



Acceptance Criteria (Definition of Done)

- ☐ All cells in `Task4_Safety_Evaluation.ipynb` execute sequentially from top to bottom without errors.
- ☐ Confidence threshold is calibrated from empirical retrieval score gaps rather than arbitrary guesses.
- ☐ Unsupported-claim detector correctly validates supported statements and flags drifted claims.
- ☐ Benchmark evaluation executes across all retrieval and safety test cases.
- ☐ Average Precision@k and Safety Pass Rate are calculated and reported with exact numbers.